HDOC DAY 65

Tower of Hanoi Using Recursion

Aim Move n disks from source rod A to destination rod C using auxiliary rod B and display all moves.

Algorithm

1. Read the positive number of disks n.

2. If n = 1, move the disk from source to destination.

3. Otherwise, move n-1 disks from source to auxiliary recursively.

4. Move the largest disk to destination.

5. Move n-1 disks from auxiliary to destination recursively.

C Program

```
#include <stdio.h>

void hanoi(int n, char from, char to, char aux)
{
    if (n == 1) {
        printf("Move disk 1 from %c to %c\n", from, to);
        return;
    }
    hanoi(n - 1, from, aux, to);
    printf("Move disk %d from %c to %c\n", n, from, to);
    hanoi(n - 1, aux, to, from);
}

int main(void)
{
    int n;
    printf("Enter number of disks: ");
    if (scanf("%d", &n) != 1 || n < 1 || n > 15) {
        printf("Enter a disk count from 1 to 15.\n");
        return 1;
    }
    hanoi(n, 'A', 'C', 'B');
    return 0;
}
```

Test Cases

Input	Expected result
1	Move disk 1 from A to C
2	3 moves
3	7 moves

Compilation and Execution (Mac Terminal)

```
clang -std=c99 -Wall -Wextra day65.c -o day65
./day65
```

Sample Output

```
Enter number of disks: 3
Move disk 1 from A to C
Move disk 2 from A to B
```

```
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
```

Important Note

Minimum moves = 2^n - 1. The input cap of 15 prevents excessively large output.

Result

The program was implemented using the required technique and produces the expected results for the given test cases.